

therefore, larger values of γ values are needed to smooth the shell structure out. The new method works equally well for calculating neutron and proton shell corrections.

We introduced a renormalized smoothing range in which the p dependence of the smoothing range was reduced considerably. Using this renormalized range the selection of the proper minimum of the plateau curves was easier.

Therefore, we recommend the use of the new procedure with finite range smoothing first of all for light nuclei, where the generalized Strutinski method can not be applied. We also recommend its use in regions being close to drip lines where the finite range smoothing seems to work somewhat better than the generalized Strutinski method.

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